

EMIR CEYLAN

Computer Science student — machine learning for healthcare

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SUMMARY

Computer Science student with hands-on experience across machine learning, full-stack software development and data science: from developing and validating models on real-world clinical datasets to building and deploying web apps with React, Docker and Firebase. Seeking a software, ML or data role in industry.

EDUCATION

Sabancı University

September 2023 - expected 2027

B.Sc. in Computer Science and Engineering · İstanbul, Türkiye

- ▶ Ranked 7,004th nationally in the YKS; 50% merit scholarship.
- ▶ Relevant coursework: Data Structures, Algorithms, Bioinformatics, Computational Biology, Machine Learning, Discrete Mathematics.
- ▶ Top 10% in Data Structures (CS204); strong grades in Algorithms (CS300), Data Science (DSA 210), Computational Biology (ENS 210).

EXPERIENCE

Machine Learning Intern

Summer 2026

Amsterdam UMC · Amsterdam, Netherlands (Erasmus+)

- ▶ Built a calibrated XGBoost classifier on MIMIC-IV v3.1 predicting 30-day post-discharge mortality for patients aged 70+ (172,575 admissions from 68,328 patients), supporting advance-care-planning triage.
- ▶ Engineered 77 clinically justified features from the 28 GB raw EHR release with DuckDB and polars.
- ▶ Reached AUROC 0.898, beating the LACE and HOSPITAL clinical scores by +0.146 and +0.066 AUROC under GroupKFold validation, with strong calibration, SHAP explainability and TRIPOD+AI reporting.
- ▶ Manuscript in preparation.

Python · XGBoost · DuckDB · polars · SHAP · scikit-learn

Undergraduate Researcher (PURE) — Human-AI Interaction in Medicine

2025 - 2026

Sabancı University · Advisor: Dr. Polat Göktaş

- ▶ Building trustworthy, explainable deep-learning models for healthcare decision support (SHAP, Grad-CAM).

Python · PyTorch · SHAP · Grad-CAM

Undergraduate Researcher (PURE) — Machine Learning for Biomedical Alloys

Ongoing

Sabancı University · Advisor: Dr. Azizeh Hosseinjany

- ▶ Developing an ML framework to predict and optimise the mechanical properties and biocompatibility of High-Entropy Alloys (HEAs) and Shape Memory Alloys (SMAs).

Python · scikit-learn · pandas

Learning Assistant — DSA 210 Data Science

Spring 2025 - 2026

Sabancı University · Supervisor: Öznur Taştan

- ▶ Mentored students in Python-based data-science workflows: exploratory data analysis and ML implementation with scikit-learn.

Python · scikit-learn · pandas · Matplotlib

Hackathon Organizer & Technical Mentor

May 2026

Deep Technologies Club, Sabancı University

- ▶ Helped organise and run Sabancı University's first hackathon; led the technical side and mentored participating teams throughout the event.

PROJECTS

Litigation Score Predictor (B2B SaaS) April 2026
ML Developer (team of 5) · ELSA Lawathon (European Law Students' Association) ·
2nd place · £20,000 prize / seed funding

- ▶ ML model that predicts litigation outcome scores and cites relevant precedent rulings (emsal kararlar) for case documents uploaded by companies and law firms, designed as a B2B SaaS product.
- Python · NLP · scikit-learn

Osteogenesis Imperfecta (OI) Diagnosis Tool September 2025 - January 2026
Lead Developer · ENS 210 Computational Biology · github.com/EmirC356/ENS210_Project

- ▶ Diagnostic tool trained on 3,000+ sequences; evaluated four ML algorithms, reaching a peak accuracy of 98.8%.
- ▶ Introduced a novel feature representation based on collagen's amino-acid structure; minimised overfitting through rigorous cross-validation.

Python · scikit-learn · Biopython

E-Commerce Platform Fall 2025
Full-Stack Developer (team of 5) · CS308 Software Engineering

- ▶ Django REST Framework backend, React/Vite frontend, PostgreSQL, JWT auth: product catalogue, cart, checkout, orders, reviews and wishlist.
- ▶ Agile collaboration via Git and Jira.

Django REST Framework · React · Vite · PostgreSQL · JWT · Git · Jira

Fitness & Accountability App Spring 2026 - present
Developer · in progress

- ▶ Calorie- and exercise-tracking app centred on friend accountability, with one-time (ephemeral) photo sharing.

This website (emirceylan.com) Spring 2024 - present
Developer · github.com/EmirCeylan356/EmirCeylanWebsite

- ▶ Originally self-hosted in Docker with persistent storage and a Firebase backend; now an Astro static site on Vercel with a Supabase-backed visitor drawing gallery.

Astro · Tailwind CSS · TypeScript · Supabase · Docker · Vercel

SKILLS

ML & DATA	XGBoost, scikit-learn, SHAP, pandas, polars, NumPy, DuckDB, SQL, Matplotlib, Biopython, Jupyter
LANGUAGES	Python, C++, JavaScript, SQL
WEB & INFRA	React, Vite, Django REST Framework, PostgreSQL, Docker, Firebase / Supabase, Astro, Git / GitHub, Jira, Ubuntu / Linux
CONCEPTS	Machine learning, Deep learning, Explainable AI, Model calibration, Full-stack web, REST APIs, Statistics, Data visualisation
SPOKEN	English (C1, fluent), Turkish (native)

CERTIFICATIONS

- ▶ UC San Diego Bioinformatics Specialization (Coursera)
- ▶ Stanford Machine Learning (Andrew Ng)

LEADERSHIP

- ▶ Board Member, Deep Technologies Club — BioTech division (2025 – present)
- ▶ President, Sabancı Archery Club (2024 – 2025)